

Forecast Evaluation

Chapter 5

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Outline

- In-sample fit vs. out-of-sample accuracy
- Loss functions
- Out-of-sample evaluation design
- The Diebold-Mariano test
- Forecast combination
- Calibration and predictive distributions

In-sample fit vs. out-of-sample accuracy

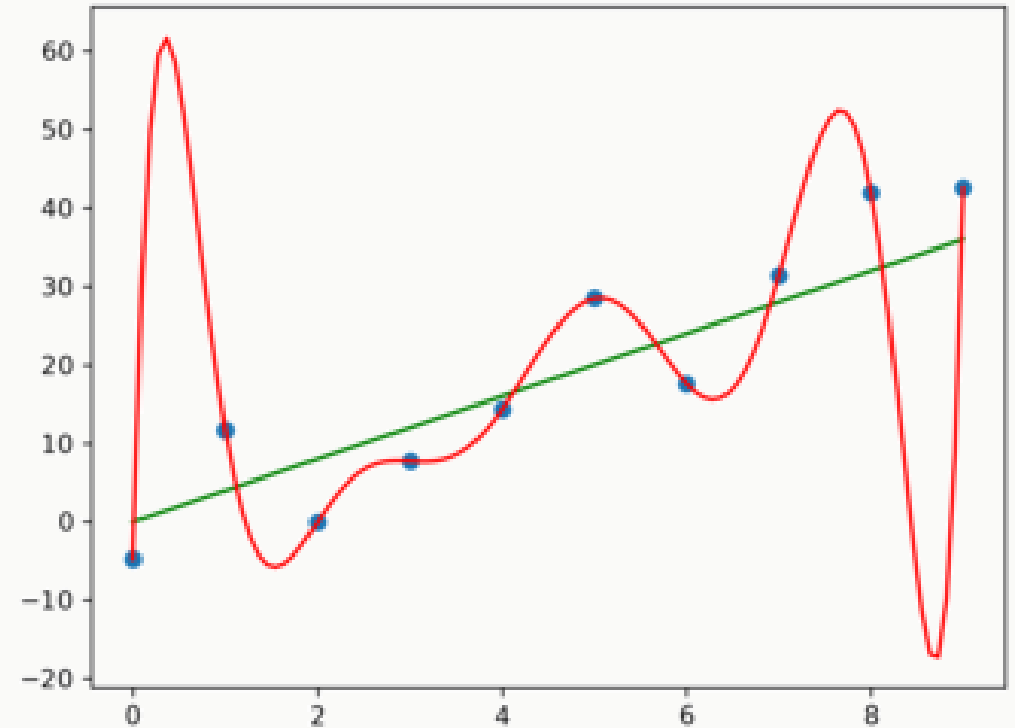
Section 1

In-sample fit vs. out-of-sample accuracy

- In-sample fit
 - How well a model explains the data it was estimated on — a measure of the model's description of the **past**
 - $\hat{y}_t$ vs. y_t when $t \in [1, \dots, T]$
- Out-of-sample-forecast accuracy
 - How well a model predicts data it has never seen — a measure of the model's usefulness for the **future**
 - $\hat{y}_{T+h|T}$ vs. $y_{T+h|T}$ when $h > 0$

In-sample fit vs. out-of-sample accuracy

- Overfitting in one plot
 - The polynomial is a **perfect** in-sample fit
 - The line is not
 - The line is still a better forecast than the perfect fit
 - **Fit** and **accuracy** are not the same



In-sample fit vs. out-of-sample accuracy

- Three gaps when using information criteria (IC) for model selection
 - Gap 1
 - IC maximizes the log-likelihood of an **assumed** distribution.
 - What if the data distribution is different than the one used to estimate the model?
 - Gap 2
 - No model can forecast unknown future structural breaks or changes in behavior (next chapter).
 - The right pre-Covid model might be wrong for the post-Covid world.
 - Gap 3
 - IC is a useful selection criterion when the dependent variable is the same across models, and all models are of the same family (i.e., SARIMAX)

Loss functions

Section 2

Loss functions

- Start defining an error (loss) to measure
 - $e_{t+h|t} = y_{t+h|t} - \hat{y}_{t+h|T}$
 - $t + h$: Forecast or target date
 - t : Forecast origin
- You should expect accuracy to decrease with increasing h

Loss functions

- “Basic” measures
 - $MAE = \frac{1}{h} \sum_{i=1}^h |e_{t+i|t}| \rightarrow$ evaluate against the median
 - $MSE = \frac{1}{h} \sum_{i=1}^h e_{t+i|t}^2 \rightarrow$ evaluate against the mean
 - $RMSE = \sqrt{MSE}$

Loss functions

- The previous measure keep y_t unit of measure
- If there are dependent variables measured in different units, then you need a unit-free measure
 - $MAPE = \frac{1}{h} \sum_{i=1}^h \left| \frac{e_{t+i|t}}{y_{t+h}} \right| \times 100$
 - Express the forecast error in percentage points of the dependent variable
 - The **MAPE Trap**: If y_t is close to zero, MAPE becomes very large
 - A problem with time series that would have values close to zero — such as interest rates

Loss functions

- Avoiding the MAPE Trap

- $$MASE = \frac{MAE}{\frac{1}{T-1} \sum_{t=2}^T |y_t - y_{t-1}|}$$

- The **Mean Absolute Scaled Error** divides the MAE by a fixed in-sample benchmark
 - If $MASE < 1$, then the model beats the naïve random walk
 - Useful when comparing dependent variables with different volatilities

Loss functions

- Asymmetric Loss: When Errors Aren't Equally Bad
 - For a central bank, inflation under 2% might be more costly than a similar but positive error
 - $\mathcal{L}(\pi - \pi^* = -1\%) > \mathcal{L}(\pi - \pi^* = 1\%)$
 - For an inventory manager, under-forecasting or over-forecasting demand can have different cost implications

- $$\mathcal{L}(e) = \begin{cases} \alpha|e|, & e < 0 \\ (1 - \alpha)|e|, & e \geq 0 \end{cases} \quad \alpha \in (0, 1)$$

Loss functions

Error e	Squared e^2	Absolute $ e $	Asymmetric ($\alpha = 0.3$)
-4	16	4	$0.3 \times 4 = 1.2$
-2	4	2	$0.3 \times 2 = 0.6$
-1	1	1	$0.3 \times 1 = 0.3$
0	0	0	0
1	1	1	$0.7 \times 1 = 0.7$
2	4	2	$0.7 \times 2 = 1.4$
4	16	4	$0.7 \times 3 = 2.8$

Out-of-sample evaluation design

Section 3

Out-of-sample evaluation design

- The **Look-Ahead Bias** problem
 - The accidental use of future information when estimating a model
 - Direct Bias
 - Estimate the model with the full sample $[1, \dots, T]$, then evaluate the forecast on a sub-sample inside T
 - Indirect Bias
 - Use a sub-sample $[1, \dots, t]$ to select a model, then evaluate on a hold-out set $[t + 1, \dots, T]$
- Data-processing
 - The data itself is constructed using out-of-sample forecasts

Out-of-sample evaluation design

- Instead, try rolling and recursive windows
 - Run both... any difference will be informative

	Rolling window	Recursive window
Window length	Fixed at W	Grows each step
Oldest observation used	$t - W + 1$	Always $t = 1$
Adapts to change	Yes — drops oldest data	Slowly — retains all data
Parameter precision	Lower (fewer observations)	Higher (more observations)
Best when	Process may be unstable	Process looks stable

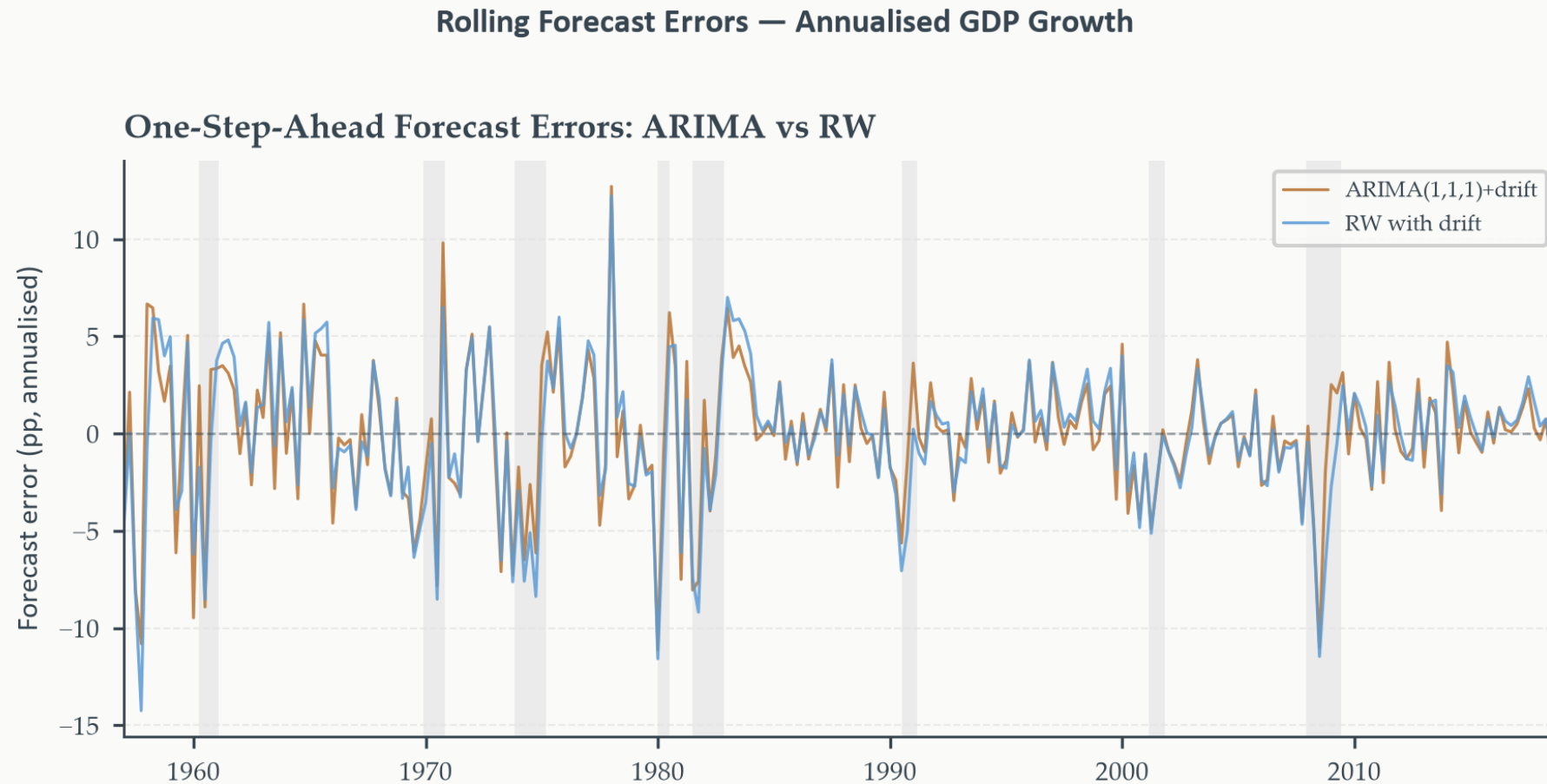
Out-of-sample evaluation design

- How to chose h and W
 - Horizon
 - Policy-dependent or based on user-need
 - A fiscal authority planning multi-year budgets will look at $h > 8$ (quarters)
 - Window
 - A trade-off
 - A smaller (larger) window provides faster (slower) adaptation to changes in data behavior but estimates are more (less) noisy

Out-of-sample evaluation design

- Evaluation approach
 - Step 1: Fit your model
 - Step 2: Collect forecast errors for $h = 1$
 - Step 3: Collect forecast errors for $h = 2$
 - Step 4: Collect forecast errors for $h = 4$
 - Step 5: Collect forecast errors for $h = 8$
 - Step 6: Compare each forecast error with its naïve model estimation counterpart

Out-of-sample evaluation design



The Diebold-Mariano test

Section 4

The Diebold-Mariano test

- Two models may have similar expected forecast accuracy, and yet provide different errors on any given finite sample by luck (random)
- The Diebold-Mariano test
 - A formal hypothesis test for equal predictive accuracy that works for any loss function
 - No assumption on parametric form
 - It does not require errors to be normally distributed

The Diebold-Mariano test

- Take the forecast error for each model for each estimate origin t
 - Step 1: calculate $d_t = \mathcal{L}(e_{1,t+h|t}) - \mathcal{L}(e_{2,t+h|t})$ for each forecast point
 - Step 2: get the average $\bar{d} = \frac{1}{n} \sum_t d_t$
 - H0: $\mathbb{E}[d_t] = 0$
 - HA: $\mathbb{E}[d_t] \neq 0$
- If model 1 is better than model 2 $\rightarrow \bar{d} < 0$
- If model 2 is better than model 1 $\rightarrow \bar{d} > 0$

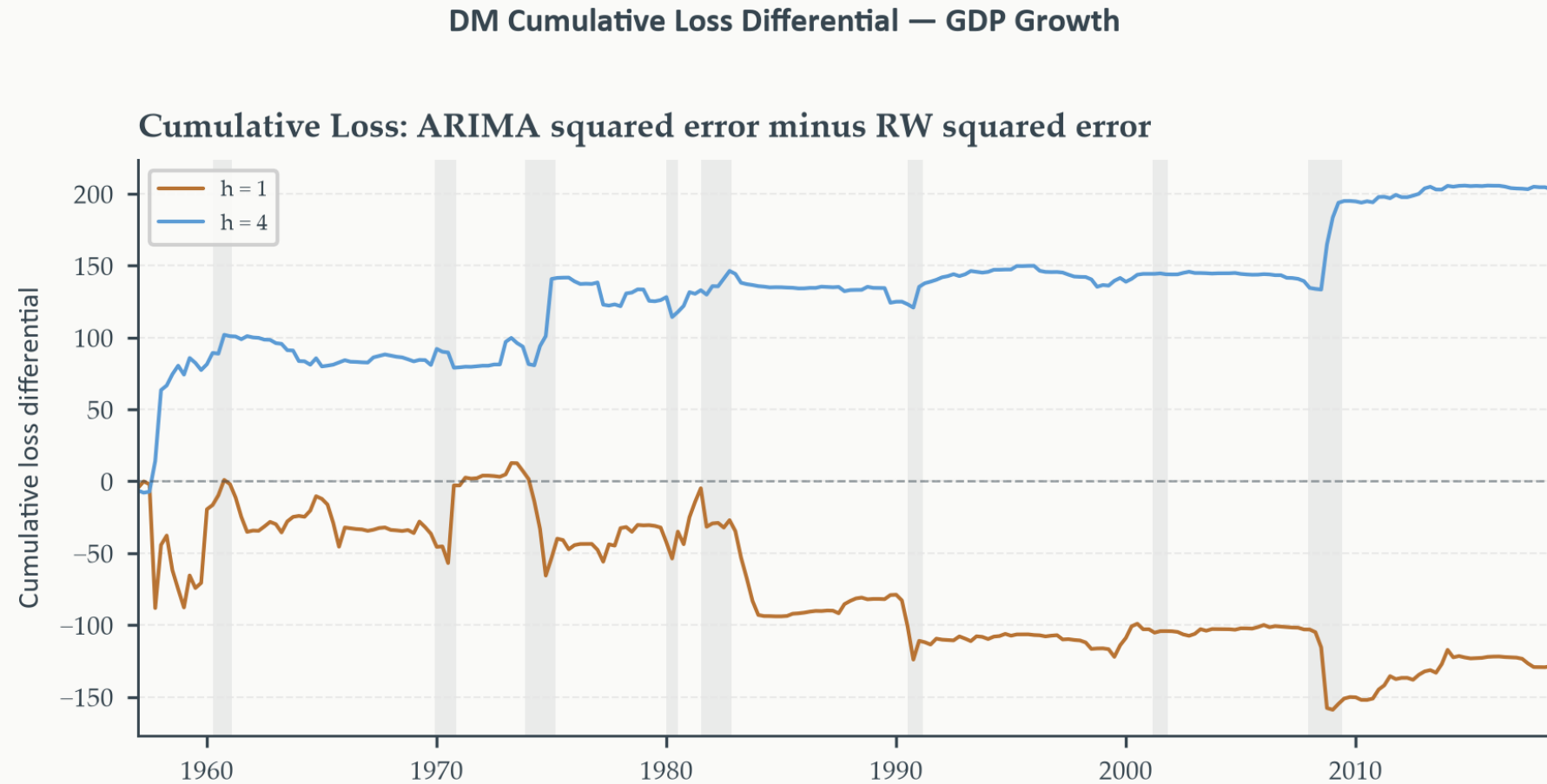
The Diebold-Mariano test

- Take the forecast error for each model for each estimate origin t
 - Step 1: calculate $d_t = \mathcal{L}(e_{1,t+h|t}) - \mathcal{L}(e_{2,t+h|t})$ for each forecast point
 - Step 2: get the average $\bar{d} = \frac{1}{n} \sum_t d_t$
 - H0: $\mathbb{E}[d_t] = 0$
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- If model 2 is better than model 1 $\rightarrow \bar{d} > 0$

The Diebold-Mariano test

- Problem, $\bar{d}/\sigma_{\bar{d}}^2$ does not have a typical distribution because errors are serially correlated: $e_{t+h|t}$ is affected by $\varepsilon_t, \varepsilon_{t+1}, \dots, \varepsilon_{t+h}$
- Solution, transform $\bar{d}$ such that has a standard normal distribution
 - Replace the variance of $\bar{d}$ with its **long-run variance** by correcting for autocovariances
 - $\hat{V}(\bar{d}) = \frac{1}{n} \left(\hat{\gamma}(0) + 2 \sum_{k=1}^K w_k \hat{\gamma}_k \right)$
 - $\hat{\gamma}_k = \frac{1}{n} \sum_t (d_t - \bar{d})(d_{t-k} - \bar{d})$
 - Newey-Bartlett weights: $w_k = 1 - \frac{k}{K+1}$ downweights distant lags
 - $DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}} \sim N(0,1)$

The Diebold-Mariano test



The Diebold-Mariano test

- Reading the DM test
 - It only says whether the forecast difference for a given forecast horizon is statistical noise or not
 - It doesn't conclude how forecasts are going to perform on non-evaluated data (the future)
 - The DM test answers a simple question: Is an observed forecast different from noise? Nothing more, nothing less

Forecast combination

Section 5

Forecast combination

- You find that one model is better than an alternative (or alternatives)
- Should you just forecast with the best model and discard the others?
 - No
 - A combined forecast can beat the best model
 - Why?
 - If errors are negatively correlated, they cancel out (just like portfolio diversification)

Forecast combination

- The combined forecast is a weighted average of 2+ forecasts
 - $y_{t+h|t}^{(c)} = \omega y_{t+h|t}^{(1)} + (1 - \omega) y_{t+h|t}^{(2)}$
 - $y_{t+h|t}^{(c)} = \omega e^1 + (1 - \omega) e^2$
- The variance of the combined error becomes
 - $VAR(e^c) = \omega^2 \sigma_1^2 + (1 - \omega)^2 \sigma_2^2 + 2\omega(1 - \omega)\sigma_{12}$
 - If $\sigma_{12} < 0$, then $VAR(e^c)$ is less than either $VAR(e^1)$ or $VAR(e^2)$

Forecast combination

- How to choose the weights?
 - Option 1: The Bates-Granger Optimal Weight
 - Option 2: Just use $\omega^* = 0.5$
 - Option 3: OLS combination

Forecast combination

- Option 1: The Bates-Granger Optimal Weight
 - Since $VAR(e^c)$ is a function of σ_1^2 , σ_2^2 , and σ_{12} , we can minimize it by choosing optimal weights
 - $$\omega^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$
 - Weight on model 1 rises when model 2's variance is high
 - If $\sigma_{12} = 0$, then $\omega^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$ (inverse-variance weighting)
 - If $\sigma_{12} = 0$, and $\sigma_1^2 = \sigma_2^2$, then $\omega^* = 0.5$

Forecast combination

- Option 2: Just use $\omega^* = 0.5$
 - Why?
 - If W (evaluation window) is short, then ω^* will be inaccurate and can degrade performance

Forecast combination

- Option 3: OLS combination
 - Regress y_{t+h} on $\hat{y}_{t+h|t}^1$ and $\hat{y}_{t+h|t}^2$
 - Allows for biased individual forecasts and general correlation structure

Forecast combination

- When to use a combined forecast?
 - The DM test fails to reject (models are comparably accurate)
 - The cumulative loss differential shows reversals (errors are canceling out)
 - Models draw on different information
- When to select one model to forecast?
 - One model consistently dominates all others

Forecast combination

RMSE ratio vs. random walk benchmark

Strategy	$h = 1$	$h = 4$
ARIMA alone	0.978	1.033
Random Walk alone	1.000	1.000
Equal weights	0.972	1.011
Bates-Granger weights	0.970 ARIMA $\omega = 0.65$	1.000 ARIMA $\omega = 0.00$
OLS combination	0.953	0.972

Calibration and predictive distributions

Section 6

Calibration and predictive distributions

- Do the forecast values fall within the forecast interval?
 - Coverage: Over n forecast origins, the fraction of times the forecast falls inside the stated interval
 - $\widehat{coverage} = \frac{1}{n} \sum_t \{y_{t+h} \in [\hat{y}_{t+h|t}^L, \hat{y}_{t+h|t}^U]\}$
- If coverage:
 - Less than forecast interval $\rightarrow$ uncertainty is understated, interval is too narrow
 - More than forecast interval $\rightarrow$ uncertainty is overstated, interval is too wide

Calibration and predictive distributions

Interval (%)	Interval (nominal)	$h = 1$	$h = 4$
90%	0.90	0.879	0.706
95%	0.95	0.931	0.794

Calibration and predictive distributions

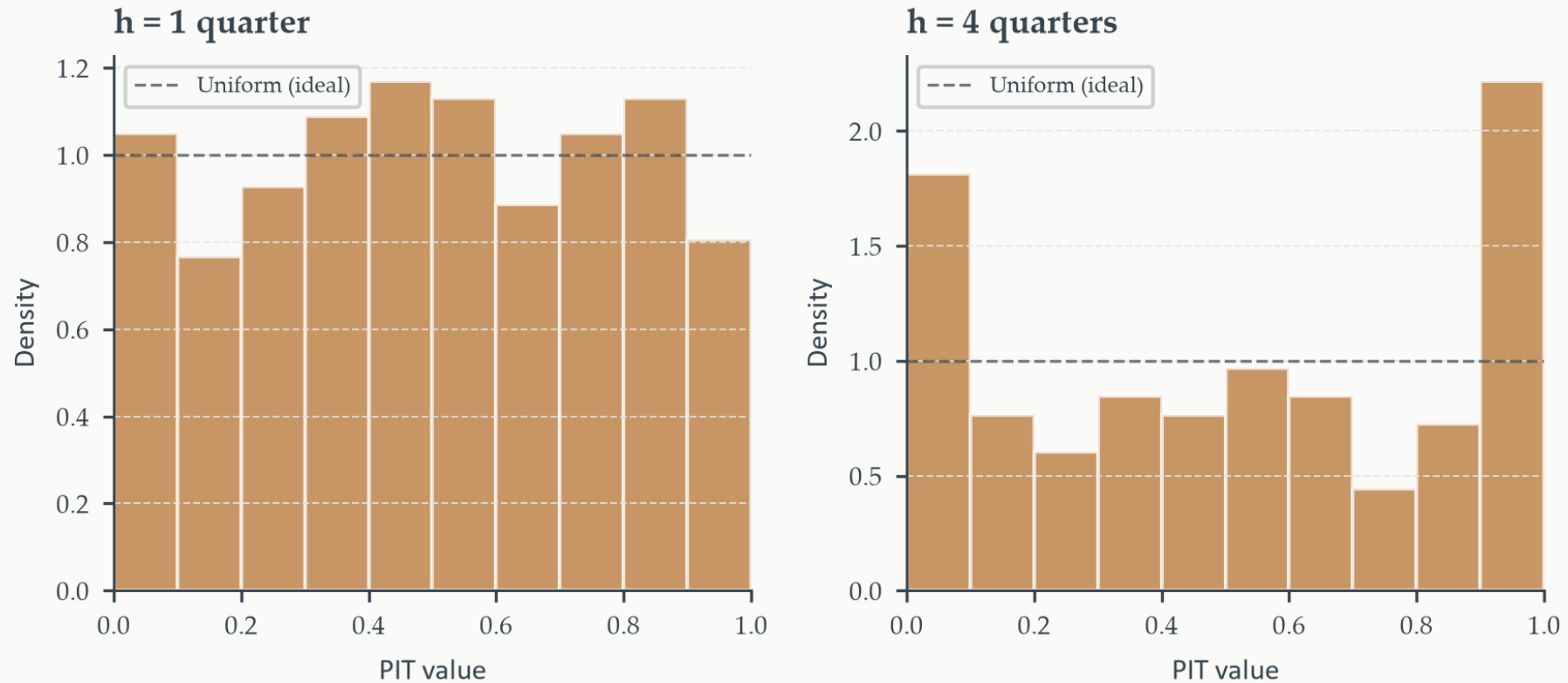
- The Probability Integral Transform (PIT)
 - Coverage looks at forecasts falling outside the interval, but is blind to the forecast distribution within the interval
 - PIT looks at the predictive distribution of $\hat{y}$ and compares with the expected distribution of $\hat{y}$

Calibration and predictive distributions

- The Probability Integral Transform (PIT) (cont...)
 - Assume a normal distribution; then
 - $z_t = \Phi\left(\frac{y_{t+h} - \hat{y}_{t+h|t}}{\hat{\sigma}_{t+h|t}}\right)$
 - Collect all z_t
 - Plot a histogram
 - Histogram shape
 - U-shape: Intervals are too narrow → the model is surprised more than it should
 - Hump-shaped: Intervals are too wide → forecast cluster tighter than the model claims
 - Horizontal: A well-calibrated model

Calibration and predictive distributions

PIT Histograms — ARIMA(1,1,1)+drift, GDP Growth



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